

This page includes reports and technical papers related to our biofouling policies.

In-water cleaning

Marine biosecurity

Vessel biofouling is a major pathway for the introduction of marine pests into Australian waters. In-water vessel cleaning and removal of biofouling is useful in minimising the biosecurity risk. However, biological waste must be captured to prevent exotic marine species establishing in our marine environment. A review of minimum viable propagule sizes of key biofouling taxa was commissioned to:

- assess the potential for release of organisms during in-water cleaning
- inform the filtration level contained in the biosecurity standard.

Download

[Assessment of reproductive propagule size for biofouling risk groups \(PDF 6.2 MB\)](#) [Assessment of reproductive propagule size for biofouling risk groups \(DOCX 176 KB\)](#)

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Chemical Contaminant Risks

Most vessels are coated with an anti-fouling coating to prevent growth of biofouling on hulls and maintain vessel efficiency. Many anti-fouling coatings contain a biocide which is released into the seawater by one of several chemical mechanisms to deter marine growth.

Reports were commissioned to help us better understand the potential chemical contaminant risks posed by in-water cleaning of anti-fouling coatings in Australian waters.

This report collates and evaluates available information on the chemical contaminant risks of in-water cleaning. It recommends ways to manage these risks.

Download

[Chemical contaminant risks associated with in-water cleaning of vessels \(PDF 6.2 MB\)](#) [Chemical contaminant risks associated with in-water cleaning of vessels \(DOCX 176 KB\)](#)

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MAMPEC modelling of key Australian ports to predict environmental impacts of in-water cleaning discharge

This report includes a Marine Antifoulant Model for Predicting Environmental Concentrations (MAMPEC). It's used to predict the amount of copper that may be released in key Australian ports under different cleaning scenarios.

Biofouling Management Requirements

The full Australian biofouling management requirements can be viewed below.

Commercial vessels

[Australian biofouling management requirements for commercial vessels \(PDF 713 KB\)](#) [Australian biofouling management requirements for commercial vessels \(DOCX 1.41 MB\)](#)

Non-commercial vessels

[Australian biofouling management requirements for non-commercial vessels \(PDF 457 KB\)](#) [Australian biofouling management requirements for non-commercial vessels \(DOCX 1.7 MB\)](#)

If you have difficulty accessing these files, visit [web accessibility](#) for assistance.

Recreational boat operators

Recreational Boat Operators Self-Management of Biofouling in Australia

A study was conducted by the Australian Bureau of Agricultural and Resource Economics and Sciences (ABARES) on recreational boat operators self-management of biofouling in Australia. [Read the summary](#) or download the full report.

Download

[Recreational boat operators' self-management of biofouling in Australia \(PDF 3.0 MB\)](#) [Recreational boat operators' self-management of biofouling in Australia \(DOCX 2.45 MB\)](#)

If you have difficulty accessing these files, visit [web accessibility](#) for assistance.

Temporal development of biofouling assemblages

In 2010, a literature review was conducted to evaluate the development of biofouling assemblages on non-toxic substrates following immersion periods of 1 – 4 weeks, in both temperate and tropical climates. We use this information to determine whether vessels undergoing hull cleaning in foreign ports may become recolonised by local biota following cleaning if they reside in these ports for a further 1 – 4 weeks prior to departure for Australia.

Download

[Temporal Development of Biofouling Assemblages \(PDF 596 KB\)](#) [Temporal Development of Biofouling Assemblages \(DOCX 624 KB\)](#)

If you have difficulty accessing these files, visit [web accessibility](#) for assistance.

Species biofouling risk assessment:

Department of Agriculture and Water Resources, February 2011

Marine biological invasions have increased throughout the world's oceans to the extent that no region is considered 'pristine'. At least 1781 species have been identified as introduced in marine or estuarine systems somewhere in the world. Many of these species have been introduced to several regions, inferring an ease of transport by human mediated mechanisms. A substantial number of these species have either demonstrable or inferred impacts.

Download

[Species biofouling risk assessment \(PDF 6.0 MB\)](#) [Species biofouling risk assessment \(DOC 5.9 MB\)](#)

If you have difficulty accessing these files, visit [web accessibility](#) for assistance.

Centre of Excellence for Biosecurity Analysis (CEBRA)

Factors that influence vessel biofouling and its prevention and management

This document describes the current scientific and technical understanding of the factors that influence biofouling and biofouling management on all vessel types, including commercial and recreational vessels. It informs and supports our policy on biofouling of international vessels entering Australian territory under the Biosecurity Act 2015 (the Act). It does not assign risk to particular vessel scenarios but is a resource that we can use to support an assessment of biosecurity risk associated with biofouling under the Act. It is a living document that will be periodically reviewed and updated as new information becomes available.

Regulation Impact Statements (RIS)

Developing mandatory biofouling requirements

In 2021, we published a decision RIS, which provided policy options for Australian Government action to improve the regulation of biosecurity risk associated with biofouling on vessels arriving into Australian territory. The Australian Government agreed with the department's recommended policy option: proactive biofouling management practices.

The Biosecurity Amendment (Biofouling Management) Regulations 2021 entered into force on 15 June 2022. From this date, the department implemented an 18-month education phase after extensive consultation with stakeholders. Further consultation and feedback was sought during the education phase.

Consultation Regulation Impact Statement

In 2019, the department consulted on a RIS which set out options for mandatory requirements to manage biosecurity risks associated with biofouling on vessels arriving in Australian territory. The submissions were considered when developing new biofouling requirements.

Source: <https://www.agriculture.gov.au/biosecurity-trade/aircraft-vessels-military/vessels/marine-pest-biosecurity/biofouling/biofouling-resources>